RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision (Delta)
Topic(s): (C) Statistics
(Questions)

1 A company makes portable speakers. Some speakers turn out to be faulty. Each carton has 24 portable speakers.

(i) State, in context, two assumptions needed for the number of faulty portable speakers in a randomly chosen carton to be well modelled by a binomial distribution. [2]

It is given that 2% of the portable speakers produced are faulty.

(ii) Find the probability that a randomly chosen carton contains at most one faulty portable speaker. [1]

(iii) A carton with more than one faulty portable speaker is considered as 'substandard'. Find the probability that, out of 50 cartons, there are no 'substandard' cartons. [2]

The company also makes flash drives packed in batches of one hundred. The probability that a flash drive being faulty is p, where $0 < p < 1$. For quality control purposes, a random sample of 3 flash drives is tested from each batch.

- If there are 2 or 3 faulty flash drives, the batch is rejected.
- If there is no faulty flash drive, the batch is accepted.
- If there is one faulty flash drive in the sample, another random sample of 3 flash drives is tested and the batch is accepted if there are no faulty flash drive in the second sample. Otherwise the batch is rejected.

(iv) Find the range of p such that we accept at least 95% of the batches based on this quality control method. [3]

2 The probability distribution for the random variable X is shown in the table.

x	-1	0	1	2
$\mathrm{P}(X = x)$	p	$\frac{1}{5}$	q	$\frac{1}{5}$

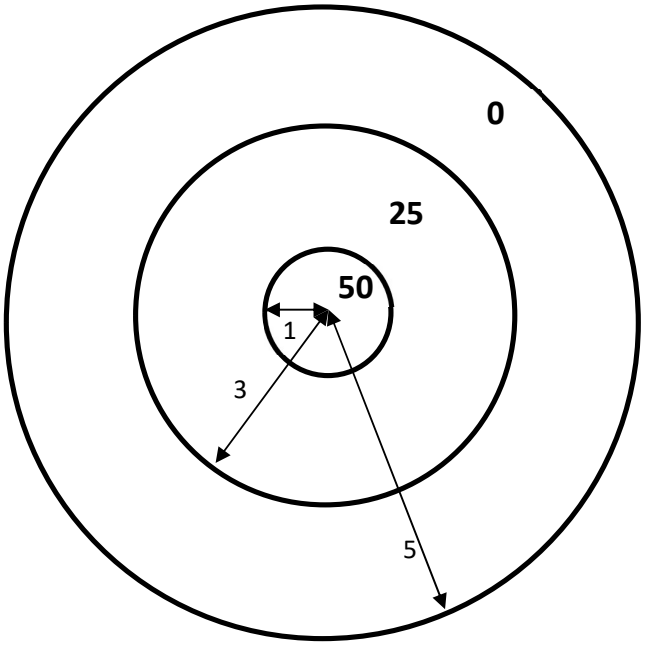
Another random variable Y is defined by $Y = 2X_1 + X_2$, where X_1 and X_2 are independent observations of X.

(i) Given that $\mathrm{P}(Y = 4) = \frac{3}{25}$, find the values of p and q. [3]

(ii) Find $\mathrm{E}(X)$ and $\mathrm{Var}(X)$. [2]

(iii) Find $\mathrm{Var}(Y)$. [2]

3 At a particular game stall, players throw darts at a target board to earn points for redeeming prizes.



The target board can be modelled by three concentric circles with radii one unit, three units and five units respectively. A player will get a score of 50, 25 or 0 points if the dart lands on the respective regions labelled **50**, **25** or **0** as shown in the diagram above.

The probability that a dart lands on any region labelled **50**, **25** or **0** is proportional to the area of the region. It can be assumed that the borders between the regions have negligible thickness and the dart will only land on the target board, which has a total area of 25π units^2.

In a game, a player throws two darts at the target board, one after the other. The first dart will be removed before the next dart is thrown. Let S be the combined score that the player achieves from the two throws.

(a) Show that $\mathrm{P}(S = 75) = \dfrac{16}{625}$. [3]

(b) Find the probability distribution of S. [4]

(c) Find the expectation and variance of S. [2]

4 The table below shows the BMI-for-age for boys aged 18 years in Singapore.

Weight Indicator / Age (years)	**Underweight**	**Overweight**
	$\leq$ 5th percentile	**$\geq$ 90th percentile**
18	≤ 16.7	$\geq m$

It is given that the BMI for 18-year-old boys follows a normal distribution $\mathrm{N}\left(22.7, \sigma^2\right)$.

(i) Show that $\sigma = 3.65$ correct to 2 decimal places. [2]

(ii) Hence, find the minimum value of BMI, *m*, such that a randomly chosen 18-year-old boy is overweight. [2]

The BMI for 15-year-old boys follows another normal distribution $\mathrm{N}\left(21.6, 3.49^2\right)$.

(iii) Find the probability that the average BMI of a randomly chosen 15-year-old boy and a randomly chosen 18-year-old boy is more than 20.1. [2]

(iv) State, in context, an assumption made in your calculation in **(iii)**. [1]

5 The mass of a "Comforta" cushion has a normal distribution with mean 170g and standard deviation 12g.

(i) Find the probability that the mass of a randomly chosen "Comforta" cushion is more than 165g. [1]

(ii) Find the probability that, of three randomly chosen "Comforta" cushions, the mass of each cushion is less than 165g. [2]

Each "Comforta" cushion comes with a cushion cover. The mass of a cushion cover has a normal distribution with mean 20g and standard deviation σ.

(iii) It is given that more than 90% of "Comforta" cushions with their covers have mass at least 170g. Find the range of values of σ. [4]

(iv) Given that $\sigma = 8$. A "Comforta" cushion costs \$0.05 per gram to produce while a cushion cover costs \$0.02 per gram to produce. A "Comforta" cushion with cover is sold at \$10 each. Find the probability that profit will be made with the sale of a "Comforta" cushion with cover. [3]

The company manufacturing "Comforta" cushions decided to launch a new product called "Serena" cushions. The mean mass of a "Serena" cushion is 250g. A sample of 50 "Serena" cushions were collected for quality check and it was found that the sample standard deviation is 17g. Find the probability that the sample mean mass of a "Serena" cushion is at most 245g. [3]

6 A chocolate manufacturer claims that the mean number of calories in a packet of chocolate is 300. A random sample of 50 packets of chocolate is selected and the number of calories, x, in each packet is measured. The results are summarised by:

$$\sum(x-300)=36, \quad \sum(x-300)^2=612.$$

(i) Find unbiased estimates of the population mean and variance. [2]

A nutritionist believes that the chocolate manufacturer has understated the mean number of calories in a packet of chocolate.

(ii) Test, at the 5% significance level, whether the nutritionist's belief is correct. [4]

(iii) Explain why there is no need for the nutritionist to know anything about the population distribution of the number of calories in the packets of chocolate. [1]

(iv) Explain, in the context of the question, the meaning of "at the 5% significance level". [1]

After receiving feedback from customers, the chocolate manufacturer improves the quality of the ingredients used. A random sample of 40 packets of the improved chocolate is taken. The mean number of calories of the 40 packets of chocolate is k and the variance is 10.

(v) Find the range of values of k such that there is sufficient evidence to conclude that the mean number of calories in a packet of the improved chocolate is not 300 at the 3% significance level. [4]

7 The amount customers spend is often correlated with the amount of time they spend in stores. According to a national survey of 500,000 shopping trips, the average time customers spend in a clothing store is 30 min. The first, second and third quartiles of the time spent, in min, are also given as follows:

Q_1	8.5
Q_2 (median)	18.5
Q_3	36.1

(a) Explain why the data suggests that shopping times do not follow a normal distribution. [1]

Jack, a clothing store manager, suspects that customers spend less time in his store than the national average.

(b) Determine whether Jack should carry out a 1-tail or a 2-tail test. State the hypotheses for the test and define any symbols you use in this context. [3]

Jack collected a random sample of 50 shopping times. The results, x (in min), are summarised as follows:

$$\sum x = 1475, \quad \sum x^2 = 45029$$

(c) Test, at the 10% significance level, whether Jack's suspicion is valid. [5]

(d) Explain what is meant by 'at the 10% significance level' in the context of the question. [1]

(e) Explain why in performing the hypothesis test, there is no need for Jack to be concerned that the shopping times of customers in his store do not follow a normal distribution. [1]

Upon closer inspection of the data, Jack realises that he has made some miscalculations. While there is no mistake in the calculation of $\sum x$, the value of $\sum x^2$ should be larger than initially calculated.

(f) Does this miscalculation affect Jack's initial conclusion? Explain your answer. [2]

❄❄❄❄❄❄❄❄ **End of Paper** ❄❄❄❄❄❄❄❄